Fraud detection model in Health Insurance Sector

3.1 Blockchain's Capabilities in Combating Insurance Fraud

Fraud Detection involves the examination and identification of irregularities and malicious behaviours within insurance claims [34]. In this regard, Table 5 summarizes how blockchain attributes can aid in this endeavour.

Type of Fraud Addressed	Characteristic	Support for insurance fraud detection
Fabricated or duplicate claims; up-coding; service unbundling.	Utilization of smart contracts for automation.	- Assistance in the identification and validation of the terms outlined in an insurance policy. - Linking various data sources through the utilization of smart contracts. - Encoding of irregular behavioural patterns.
Billing for services not rendered; identity theft; unauthorized claims.	Implementation of digital authentication and access regulations.	- Digital keys are safely kept in digital wallets. - Incorporation of access rules and digital authentication procedures into smart contracts.
Collusive networks; fast-track claims abuse; overutilization	Integration of collective intelligence.	- Peer-to-peer platforms foster multiple points of collaboration, covering areas like claims, fraud detection, ensuring that insured party payments are in compliance, control of hospitalizations, and frequency of prescriptions.
Data tampering; price manipulation; illicit kickbacks	Assurance of data transparency and immutability.	- Preservation of the integrity of healthcare records, rendering them tamper-proof. - Oversight of medications dispensed by pharmacists - Reinforcement of the Know Your Customer (KYC) and Anti-Money Laundering (AML) processes.
System tampering; denial-of-service; reputation manipulation	Adoption of decentralized systems.	- Utilization of distributed storage via peer-to-peer networks. - Ensuring that a system or service is consistently accessible, dependable, and can continue functioning even in the presence of errors or issues. - Assistance in the functioning of reputation score systems.

Table 5. In what ways can blockchain aid in identifying instances of healthcare insurance fraud?

3.2 Instance Identification of healthcare insurance fraud

Prior to detailing the architecture of our model, it is important to clarify how fraud is identified within the proposed framework. The model focuses on the detection of common fraud typologies in health insurance, such as duplicate claims, billing for unprovided services, and identity manipulation. These deceptive practices often exploit the lack of transparency and interoperability among systems and stakeholders. To address this, our solution leverages blockchain’s key properties (immutability, decentralization, and automation through smart contracts) to create a shared infrastructure where every entity involved in the insurance process contributes to fraud prevention. By integrating encrypted data access logs, standardized policy and claim tokens, and verified health records, the model facilitates real-time validation and cross-referencing of information. This allows for the early detection of suspicious patterns and ensures that each step in the insurance process (from prescription issuance to claim settlement) is transparent, traceable, and resistant to manipulation.

This section introduces a multi-layered model designed to detect fraudulent activity in health insurance. As illustrated in Figure 3, the model is structured into four interconnected layers, each responsible for a specific set of functions within the fraud detection framework.

To begin with, there's the P2P mining pool layer. This serves as the fundamental foundation, responsible for establishing, upholding, and executing the protocol to guarantee the trustworthiness and dependability of the system's functions. In other words, it ensures the consensus process.

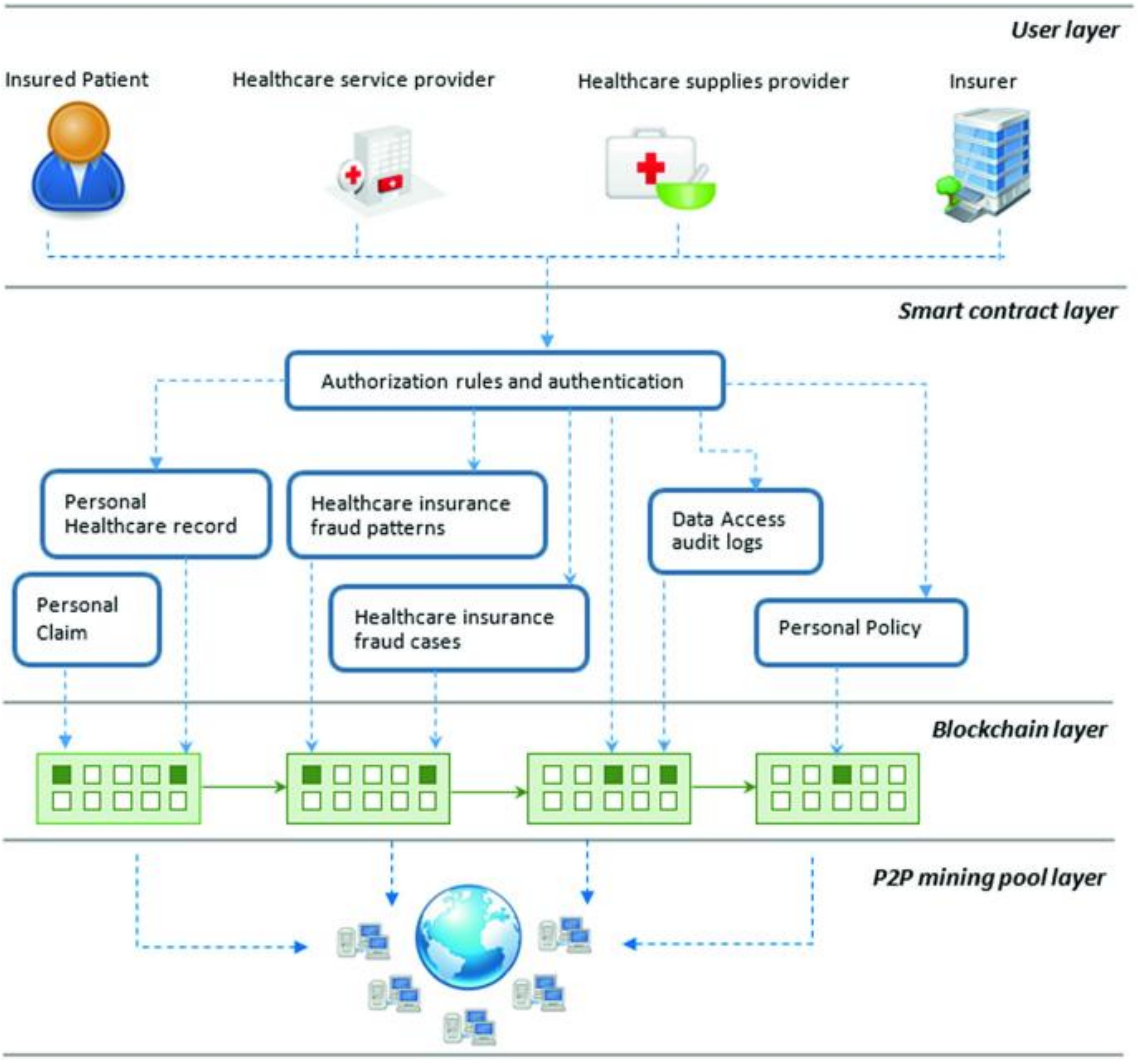


Figure 3 A hierarchical summary for identifying healthcare insurance fraud through blockchain

The second layer, the blockchain layer, is where transaction blocks are securely added in a sequential manner over time. These transactions can represent digital tokens, including cryptocurrencies and smart contracts. The arrangement and management of these tokens are orchestrated by the Layer for Peer-to-Peer Mining Pools.

The third layer is the smart contract layer, where specific functionalities within smart contracts play a crucial role in autonomously detecting fraud. In this context, seven smart contracts are employed to facilitate the process:

- Authorization and authentication rules. This layer acts as the primary entry point for end-users into the system, serving as the access interface. Here, the digital identities of users are verified, ensuring that they gain appropriate access to data based on the permissions they were granted in advance. Each entity has the flexibility to modify its authorization rules as needed.
- Data access audit logs. Access to the blockchain structure is recorded, and each access is encrypted and linked to the user entity. Furthermore, each access is recorded in an external reference database.

- Health insurance fraud cases. Deceptive claims are officially recorded, and each instance of fraud is associated with the fraudulent party.
- Fraud patterns in health insurance. To mitigate fraud, collaboration among different stakeholders is essential. The necessary data might be dispersed across diverse databases. Consequently, the use of a smart contract based on blockchain becomes crucial to tackle issues related to interoperability. A smart contract is designed to allow the registration of data references, guaranteeing the security of data from both internal and external origins. In this framework, a decentralized anti-fraud service is established, which implements and links approaches, types, and malicious behavioural patterns based on expert recommendations from external sources (Figure 3). To accomplish this, these descriptions can be encoded within the blockchain or in external systems. This can be achieved through the application of clustering algorithms and regression analysis [31], data mining [32], and decision tree development.
- Personal health record. Blockchain keeps a record of the medical prescriptions and diagnosed illnesses of each covered patient. Every token is linked to the identity of the insured person. The insured person (using a token) has the ability to grant permission for the access or sharing of their personal health records with various entities. This data may also be stored in locations external to the blockchain. In such instances, each token acts as a pointer or reference to specific records held in other external databases. This approach helps resolve compatibility problems that can arise between different systems.
- Personal policy. The automated and documented contractual agreement between the insured person and the insurance provider outlines the benefits, policy duration, limitations, and exclusions related to the covered events or contingencies.
- Personal claim. Every claim in the blockchain is securely documented. A claim is represented as a token connected to the insured person, the insurance company, and the healthcare service provider. Current details about the claim's status are available. This intelligent contract streamlines the automatic settlement process among the relevant stakeholders.

Finally, a user layer. This illustrates the five ultimate entities in the framework. Figure 4 offers a functional explanation of the model.

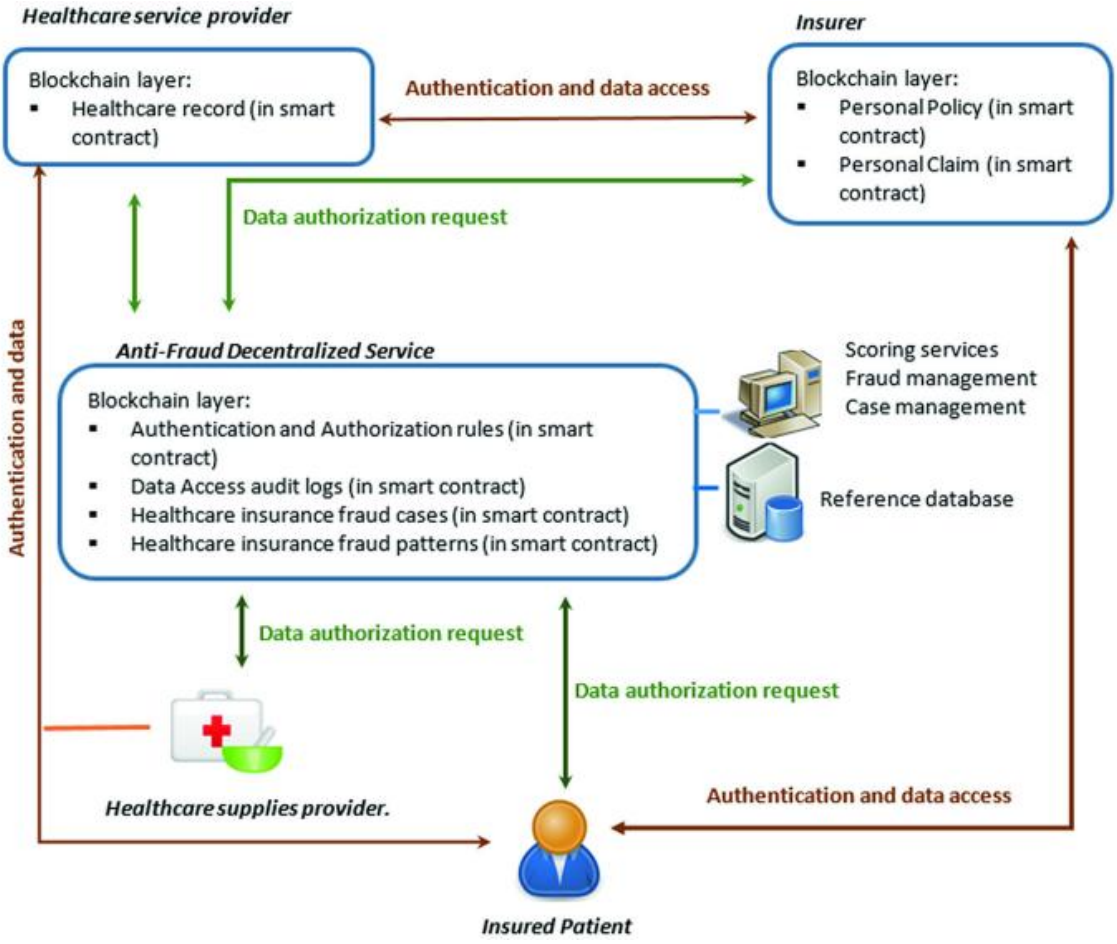


Figure 4. Blockchain-driven approach for identifying fraud in health insurance

- Patient: This is the policyholder who provides authorization for the reading of their medical history. This permission is granted through a healthcare service provider utilizing an anti-fraud service.
- Insurer: This entity is responsible for maintaining and updating the insurance policies of the insured individuals. Consequently, it provides and restricts access to this information by adhering to authorization and authentication guidelines configured within an anti-fraud service.
- Healthcare Service Provider: This entity is tasked with recording and preserving patients' health records, including medical diagnoses and prescriptions. It also facilitates access to these records through authorization and authentication mechanisms configured within an anti-fraud service.
- Healthcare Supplies Provider: These entities include pharmacies and other providers of medical supplies. Their primary function is to dispense medications in accordance with medical prescriptions, which are documented in the patient's medical records.
- Decentralized Anti-Fraud Service: This service operates within the blockchain layer and performs two main functions through the use of smart contracts. Firstly, it sets up authentication and access rules. Its role is to define all the guidelines for accessing blockchain records, which include healthcare records, insurance policies, fraud cases, fraud patterns, and individual claims. Secondly, it establishes collaboration rules with other external services. Points for validating information from external sources and settling claims can be recorded through a smart contract. In this context, collaborating with other services focused on fraud management and scoring is highly significant. Hence, the prior research, cases, patterns, and methodologies for detecting fraud in health insurance can be integrated into the framework presented in Figure 3.